

Level : BDS

Exam : BDS I (new course)

Time : 2 hour 30 minutes

Date : October 24, 2021

Subject: Biochemistry

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Discuss diagrammatically the absorption of non-heme iron, its transport and storage. How does hepcidin regulate iron absorption? $\rightarrow$ by binding to and degrading the hepcidin protein which prevents being released into bloodstream. [4+2=6] *ferropotro*
2. Illustrate with diagram the Watson and Crick model of DNA and write down its salient features. [3+2=5]
3. Write down the irreversible steps of glycolysis. Add a note on rapaport leubering cycle and its significance. [3+2=5]
4. Write short notes on:
a. Glycosaminoglycans $\rightarrow$ done (62)
b. Glycolytic products in RBCs
c. Sickle-cell haemoglobin and its laboratory diagnosis

TCA is ... because it is the final pathway where oxidation of carbohydrates & protein converges to produce energy.
① Convergence of metabolic pathway
 $\rightarrow$ Glucose is metabolized through glycolysis to pyruvate + to A-CoA
 $\rightarrow$ fat $\rightarrow$ ACoA
 $\rightarrow$ Protein $\rightarrow$ Acetyl CoA
② Central role in energy production
oxidizes A-CoA to NADH, FADH₂ & GTP
③ Link to ETC
④ Intermediates [4+2=6] for biosynthesis

SECTION "C" [25 marks]

5. Write down the steps of TCA cycle. Explain why it is called the common terminal metabolic pathway. [4+2=6] *for biosynthesis*
6. List the components of electron transport chain. Explain the chemiosmotic theory. [2+3=5]
7. Define β -oxidation. How is palmitic acid oxidized inside mitochondria? Write the fate of acetyl CoA. 9ul. [1+2+2=5]
8. Write short notes on: [3 $\times$ 3=9]
a. Fat soluble vitamins
b. Regulation of cholesterol biosynthesis.
c. Free oxygen radicals

KATHMANDU UNIVERSITY
2015

Subject: Biochemistry

Level : BDS

Paper : I

Exam : BDS I (new course)

Marks : 50

Time : 2 hr. 30 minutes

Date : September 11, 2015

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. a. Explain the mechanism of competitive inhibition of an enzyme with the help of a figure. [3]
b. What are isozymes? Explain with an example. [1+2=3]
2. Mention the genetic defect in sickle cell anemia. How does this lead to sickling of the RBC? [1+4=5]
3. What is the biochemical function of vitamin B₁₂? How does its deficiency lead to pernicious anemia? [1+4=5]
4. Write short notes on: [3×3=9]
 - a. Scurvy
 - b. Organophosphorous poisoning
 - c. Phospholipids

SECTION "C" [25 marks]

5. Explain with the help of a figure the chemiosmotic theory of oxidative phosphorylation. Explain the action of uncouplers with an example. [4+2=6]
6. Describe the role of creatine phosphate in the heart. Explain calcium induced calcium release in the cardiac myocyte. State the role of calcium in cardiac muscle contraction. [2+2+1=5]
7. Explain the role of LDL in the generation of atherosclerotic plaque. Mention two ways to lower blood cholesterol. [3+2=5]
8. Write short notes on: [3×3=9]
 - a. Reactive oxygen species
 - ⑥ Respiratory burst
 - c. Cardiac markers

Level : BDS

Exam : BDS I

Time : 3 hour

Date : October 19, 2016

Subject: Biochemistry

Paper : I

Marks : 70

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "A" [35 marks]

1. Define and classify polysaccharides with examples. Write down four major biomedical functions of heteropolysaccharides. (0.5) [2+3=5]
2. Describe structural organization of protein with examples. [4+1=5]
3. Define and classify lipids. List out the importance of phospholipids. (2) [3+2=5]
4. Write down the differences between DNA and RNA. 23 [5]
5. Define cofactor and coenzyme. Give four examples of coenzymes. [3+2=5]
5. Define fat soluble and water soluble vitamins? Write down the dietary source and deficiency symptoms of Vitamin B₁₂. [1+1+3=5]
1. Enlist four major elements and four trace elements. Write down the dietary sources, daily requirement and functions of magnesium. [2+1+1+1=5]

SECTION "B" [35 marks]

- Explain with reactions, breakdown of glucose to pyruvate [5]
- Describe iron metabolism and its regulation with the help of a diagram. [5]
0. Sketch a pathway of haem biosynthesis. [5]
1. What is oxidative phosphorylation? Illustrate the electron transport chain with an explanatory diagram. [1+4=5]
2. Describe digestion and absorption of fat in detail. [3+2=5]
13. Write short notes on: [2×5=10]
 - a. Salvage pathway
 - b. Free radicals and antioxidant enzymes

Subject: Biochemistry

Paper : I

Marks : 50

Level : BDS

Exam : BDS I (new course)

Time : 2 hour 30 minutes

Date : October 19, 2016

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Describe the primary, secondary and tertiary structures of proteins. What are the forces, which stabilize them? [6]
2. What are the differences between competitive and non-competitive inhibition? Give two examples for competitive inhibition. [5]
3. What is porphyria? Classify different types of porphyrias. Give an account of acute intermittent porphyria. [5]
4. Write short notes on: [3×3=9]
 - a. Regulation of iron absorption
 - b. Significance of HMP shunt pathway
 - c. Synthesis of catecholamines

SECTION "C" [25 marks]

5. Write the steps of citric acid cycle. Explain why citric acid cycle is also known as amphibolic cycle. [6]
6. With a simple illustration, describe how endogenously synthesized lipids are transported to different tissues. [5]
7. Name enzymes that are used in the diagnosis of myocardial infarction. Explain the pattern of rise following acute myocardial infarction. [5]
8. Write short notes on: [3×3=9]
 - a. Chemiosmotic theory of oxidative phosphorylation
 - b. Anti-oxidants
 - c. Reactions of ketone bodies synthesis

BDS
BDS I (new course)
2 hour 30 minutes
January 2, 2017

Subject: Biochemistry

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

Answer each section in a separate answer book
Write brief, relevant and legible answers
Illustrate your answer with well labeled diagrams whenever required
All questions are compulsory

SECTION "B"
[25 marks]

Define carbohydrate. Classify them with example. Mention the biological role of hetero polysacchrides. [1+3+2=6] 59

Sketch a pathway of Heme synthesis. Add a note on porphyria. [3+2=5]

Velocity of an enzyme catalyzed reaction can be affect by various factors explain them. Add a note on Km value of an enzyme. [4+1=5]

Write short notes on:

[3×3=9]

1. Sulphur containing amino acid and their role in protein structure.
2. Pathway of Catecholamines synthesis
3. Iron deficiency anaemia

SECTION "C"
[25 marks]

What are lipoproteins? Explain the metabolism of VLDL with label diagram. Add a note on reverse cholesterol transport. [1+3+2=6]

Draw a well label diagram of Electron Transport chain (ETC). Discuss about oxidative phosphorylation. [3+2=5]

What is β -oxidation? Write down the activation, transport and oxidation of palmitic acid in mitochondria. [1+4=5]

Write short notes on:

[3×3=9]

1. Cardiac markers
2. Free radical and antioxidants
3. Importance of HMP shunt pathway

Subject: Biochemistry

Level : BDS

Exam : BDS I (new course)

Time : 2 hour 30 minutes

Date : December 25, 2017

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Classify lipids on the basis of number of carbon atom and double bond present. Write down the importance of phospholipids in our body. [2+3=5]
2. a. Define coenzymes. Give an example of a reaction requiring the coenzyme form of Nicacin. [1+2=3]
b. List six functions of proteins in a living organism. [3]
3. Describe iron metabolism with suitable diagram. [5]
4. Write short notes on: [3×3=9]
 - a. Organophosphorous poisoning
 - b. Thalassemias
 - c. Iron deficiency anemia

SECTION "C" [25 marks]

5. Draw the electron transport chain showing its location in the cell. Describe how the chemiosmotic theory explains the synthesis of ATP. [3+3=6]
6. Describe VLDL metabolism with a self explanatory diagram. [5]
7. Draw the pathway for synthesis of palmityl CoA starting with acetyl CoA. [5]
8. Write short notes on: [3×3=9]
 - a. Apoprotein
 - b. Atherosclerosis
 - c. TCA cycle

d. An enzyme is a protein that functions as a catalyst to mediate & speed a chemical reaction.

• Coenzyme is a substance that enhances the action of an enzyme.

Level : BDS

Exam : BDS I (new course)

Time : 2 hour 30 minutes

Date : July 9, 2018

Subject: Biochemistry

Paper : 1

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Define and classify carbohydrate with suitable examples. [5]
2. Classify amino acid and write down the functions of proteins. [5]
3. Define enzyme, co-enzyme and isoenzyme with suitable examples. [6]
4. Write short notes on: [3×3=9]
 - a. Regulation of iron metabolism
 - b. Phospholipid
 - c. Folate trap

SECTION "C"
[25 marks]

5. Describe chylomicron metabolism with illustration. [6]
6. Write down the steps of beta-oxidation of fatty acid. [5]
7. Draw a diagram of ETC showing the components. [5]
8. Write short notes on: [3×3=9]
 - a. Uncouplers of ETC
 - b. Functions of cholesterol
 - c. Lipid profile test

KATHIMANDU UNIVERSITY
December 2018/January 2019

Level : BDS

Exam : BDS I (new course)

Time : 2 hour 30 minutes

Date : January 3, 2019

Subject: Biochemistry

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Define and classify lipids. Mention the important biological functions of phospholipids. [1+3+2=6]
2. Define glycolysis. Write down and explain the three irreversible steps of glycolysis. [1+4=5]
3. Write down the absorption, transport, storage and release of iron in our body. [5]
4. Write short notes on:
a. Dopamine synthesis
b. Porphyria: definition with one example
c. Heteropolysaccharides: definition with two examples [3×3=9]

SECTION "C"
[25 marks]

5. Write down the important steps of cholesterol biosynthesis. [6]
6. Describe Krebs's cycle with an explanatory diagram. [5]
7. What are ketone bodies? Describe their formation and utilization. [5]
8. Write short note on:
a. Composition and function of HDL
b. Cardiac markers
c. Biochemical functions of eicosanoids [3×3=9]

KATHMANDU UNIVERSITY
February/March 2020

Level : BDS

Subject: Biochemistry

Exam : BDS I

Paper : I

Time : 2 hour 30 minutes

Marks : 50

Date : February 28, 2020

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

Section 'B'
[25 marks]

1. Define carbohydrates. Classify carbohydrates with examples. [1+5=6]
2. Define protein and explain its structural organization with suitable diagram. [1+3+1=5]
3. What is enzyme? Describe the enzyme inhibition. [1+4=5]
4. Write short notes on: [3×3=9]
 - a. Porphyria
 - b. Essential fatty acids
 - c. Hierarchy of protein structure

Section 'C'
[25 marks]

5. What are apo-lipoproteins? Classify lipoproteins. Illustrate the metabolic cycle of chylomicrons. [1+2+3=6]
6. What are ketone bodies? Explain the reaction leading to the formation of them. [1+4=5]
7. Give a diagrammatic representation of citric acid cycle. [5]
8. Write short notes on: [3×3=9]
 - a. Anti-oxidants and their role in biological systems
 - b. Cardiac markers and their role in diagnosis of Myocardial infarction
 - c. Chemi-osmotic theory of oxidative phosphorylation.

KATHMANDU UNIVERSITY
March/April 2021

Level : BDS

Exam : BDS I

Time : 2 hr. 30 minutes

Date : March 24, 2021

Subject: Biochemistry

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Define, classify and write down the function of carbohydrate, lipid and amino acids. [2+2+2=6]

2. What are the differences between competitive and noncompetitive inhibition? Name two therapeutic use of competitive inhibition of enzyme. [4+1=5]

3. Define neurotransmitters. Describe the synthesis and degradation of acetylcholine. [1+4=5]

4. Write short notes on:

[3×3=9]

- Regulation of heme synthesis.
- Homopolysaccharides.
- Glycolysis in the RBC.

SECTION "C"
[25 marks]

1. What are ketone bodies? Illustrate the pathway of ketone bodies synthesis and utilization. [1+3+2=6]

2. What do you mean by oxidative phosphorylation? Explain the chemiosmotic theory with necessary diagrams. [1+4=5]

3. Define biological oxidation and explain chemiosmotic theory. [2+3=5]

4. Write short notes on:

[3×3=9]

- List antioxidant systems.
- Chylomicrons.
- Cardiac markers.

KATHMANDU UNIVERSITY
August/ September 2021

Level : BDS

Subject: Biochemistry

Exam : BDS I (new course)

Paper : I

Time : 2 hour 30 minutes

Marks : 50

Date : August 27, 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"

[25 marks]

1. Classify proteins on the basis of functions with suitable examples. Write down the structural organization of secondary proteins. [3+3=6]
2. Define enzyme, coenzyme, and isoenzyme. Explain factors that affect enzymes activity. [2+3=5]
3. Define neurotransmitter, write down the criteria to be considering a substance as neurotransmitters and classify them. [1+2+2=5]
4. Write short notes on: [3×3=9]
 - a. Reaction of glycolysis with rate limiting enzymes.
 - b. Thalassemia.
 - c. Myasthenia gravis.

SECTION "C"

[25 marks]

5. Sketch the explanatory diagram of TCA cycle. Explain why TCA cycle is known as amphibolic pathway. [4+2=6]
6. What are ketone bodies? Name the physiological and pathological conditions in which synthesis of ketone bodies are increased. [1+2+2=5]
7. What do you mean by oxidative phosphorylation? Explain the chemiosmotic theory with necessary diagrams. [1+4=5]
3. Write short notes on: [3×3=9]
 - a. HDL metabolism.
 - b. Regulation of Fatty acid synthesis.
 - c. Functions of eicosanoids.

KATHMANDU UNIVERSITY
October/November 2021

Level : BDS

Subject: Biochemistry

Exam : BDS I (new course)

Paper : I

Time : 2 hour 30 minutes

Marks : 50

Date : October 24, 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"

[25 marks]

1. Discuss diagrammatically the absorption of non-heme iron, its transport and storage. How does hepcidin regulate iron absorption? [4+2= 6]
2. Illustrate with diagram the Watson and Crick model of DNA and write down its salient features. [3+2=5]
3. Write down the irreversible steps of glycolysis. Add a note on rapaport leubering cycle and its significance. [3+2=5]
4. Write short notes on: [3×3= 9]
 - a. Glycosaminoglycans
 - b. Glycolytic products in RBCs
 - c. Sickle-cell haemoglobin and its laboratory diagnosis

SECTION "C"

[25 marks]

5. Write down the steps of TCA cycle. Explain why it is called the common terminal metabolic pathway. [4+2=6]
6. List the components of electron transport chain. Explain the chemiosmotic theory. [2+3=5]
7. Define β -oxidation. How is palmitic acid oxidized inside mitochondria? Write the fate of acetyl CoA. [1+2+2=5]
8. Write short notes on: [3×3=9]
 - a. Fat soluble vitamins
 - b. Regulation of cholesterol biosynthesis.
 - c. Free oxygen radicals